Deep-Learning-Based Object Detection and Tracking of Debris Flows in 3-D Through LiDAR-Camera Fusion

Jacob Hirschberg, Valentin Tertius Bickel, and Jordan Aaron

***Abstract*—Debris flows are mixtures of water and sediments and are a significant natural hazard in mountainous regions. A novel monitoring approach combining LiDAR sensors and video cameras enables detailed observations of debris-flow dynamics, achieving high spatial (<2 cm) and temporal (10 Hz) resolution in the field. This technology allows for analyzing individual objects within the flow in unprecedented detail, essential for understanding debris-flow behavior. However, generating large datasets and enabling real-time monitoring requires automated detection and tracking of fast-moving debris-flow features. In this study, deep-learning algorithms for object detection (YOLOv5 and YOLOv8) were applied to optical images to identify orders of magnitude more debris-flow features than with existing approaches, such as surge waves (SWs) and boulders. Projecting these detections onto the LiDAR point clouds enabled the extraction of kinematic and geometric properties, including velocity and object size. The surge-wave detector performed exceptionally well, with mean average precisions (mAP) above 0.9. While smaller features like boulders and wood were more difficult to detect, the small feature (SF) detector's mAP still exceeded 0.7. Testing the detectors at different channel locations showed that a multidomain detector can reduce labeling efforts by over 75% while maintaining performance. We further demonstrated that SWs accelerate smaller features like wood, and used the dataset to calculate a wave-breaking index (adapted from coastal engineering) with a threshold of 0.4. The developed methods represent a step forward in automated debris-flow monitoring, offering scalable tools for real-time hazard assessment and deeper insight into debris-flow dynamics.**

***Index Terms*—Debris flow, deep Learning, LiDAR, natural hazard, obejct detection, object tracking.**

I. Introduction

DEBRIS flows are extremely rapid (>5 m/s) surging flow-like landslides that occur periodically in steep mountain channels [1]. They are a natural hazard type that causes an average of 1275 fatalities per year, which accounts for about half of casualties among all types of landslides [2], [3]. Furthermore, debris flows regularly damage infrastructure, property, agricultural land, and managed forests [4]. Debris flows also act as an important geomorphic process, as they shape landscapes by transferring substantial amounts of sediment from hillslopes to the downstream river system and by incising into bedrock as they flow [5], [6], [7]. Understanding the dynamics of debris flows is thus critical for reducing losses from natural hazards and furthering our understanding of sediment transfer processes in alpine landscapes.

Debris-flows are often treated as a continuum e.g., [8], [9]; however, it has long been known they are composed of a mixture of coarse and fine grained particles, wood, and water, whose interaction results in their high destructive and erosional potential [10], [11], [12]. Debris flows can feature a steep and unsaturated bouldery front, which moves slower than the liquefied slurry behind, e.g., [12], [13]. Theoretical and experimental work suggest that the bouldery front is the product of longitudinal sorting, whereby larger boulders tend to be transported to the front [14], [15]. As the bouldery front moves slower than the liquefied part behind, boulders can be deflected or overridden and recirculated [16], [17]. The interaction of coarse particles with the liquefied slurry further governs other processes that are characteristic of debris flow motion, such as levee formation and self-channelization. These latter mechanisms hinder lateral spreading of the flow and lead to the long runout [18].

A unique yet diagnostic feature of debris flows is their surging behavior [1], [11]. Surges are challenging both from the scientific and practical perspectives as they magnify the peak discharge and impact pressures, and therefore hazard potential [19], [20], [21], [22], [23]. These surges are generated by different mechanisms such as temporary storage in more gently sloped parts of the channel with subsequent release [24], longitudinal sorting [18], and "surge waves (SWs)" which spontaneously grow out of small surface instabilities [19]. Laboratory and theoretical experiments have shown that SWs play an important role in particle segregation [23], but the data to confirm this in the field are yet missing.

Our understanding of these fundamental debris-flow mechanisms is severely limited by a lack of direct field observations of these phenomena. Several field sites to study in situ debris flows have been instrumented (see [25] for a summary). However, commonly used sensors such as radar or laser point measurements, cameras, or seismic sensors are limited in their spatial and/or temporal resolution. This restricts their applicability for studying the 3-D movement vectors of indi-

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Jacob Hirschberg and Jordan Aaron are with the Engineering Geology, ETH Zurich, 8092 Zürich, Switzerland, and also with Swiss Federal Research Institute for Forest, Snow and Landscape Research WSL, 8903 Birmensdorf, Switzerland (e-mail: jacob.hirschberg@eaps.ethz.ch).

Valentin Tertius Bickel is with the Center for Space and Habitability, University of Bern, 3012 Bern, Switzerland.

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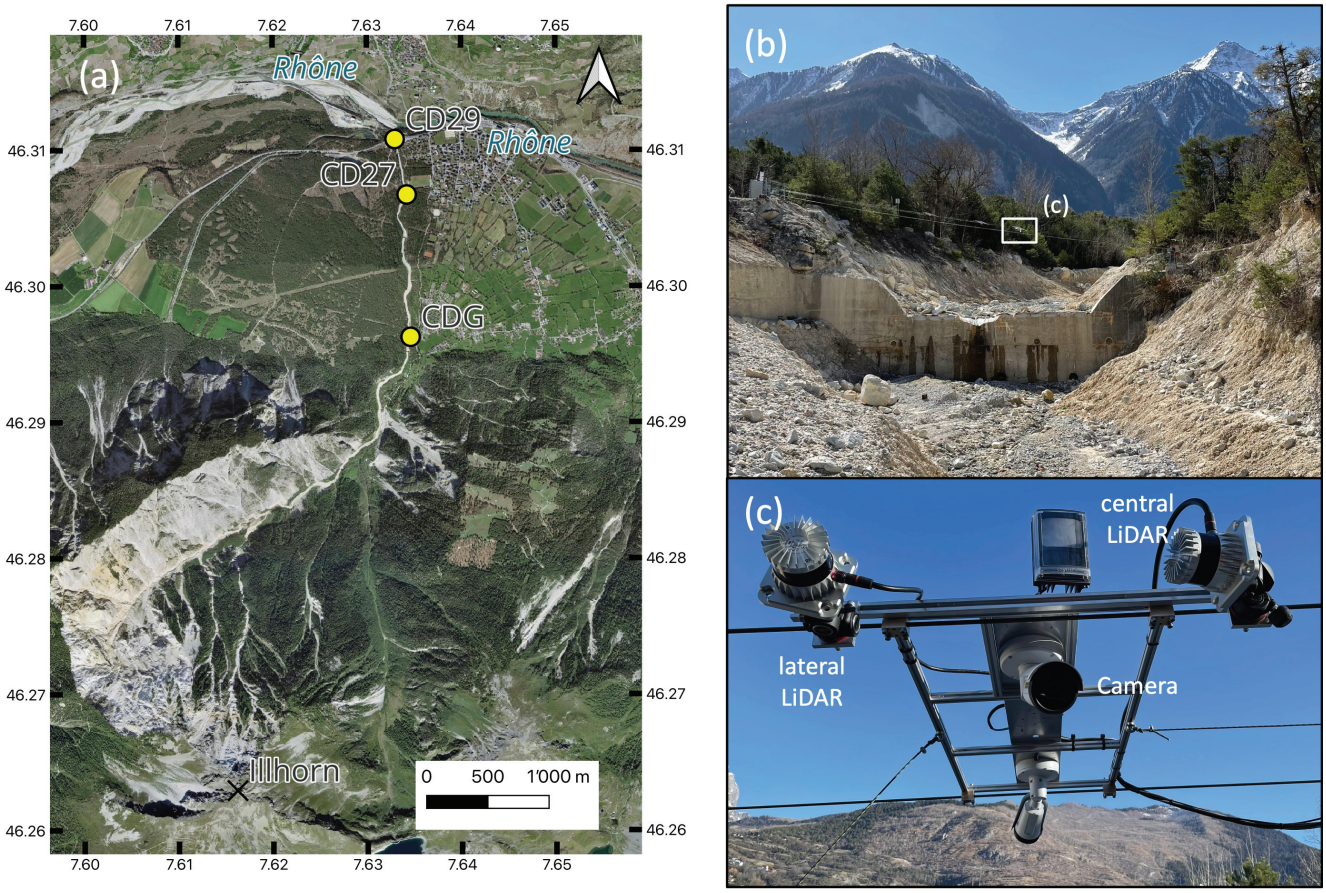


Fig. 1. Overview of the study site and monitoring stations. This study relies on data from monitoring stations at three CDs (CDG, CD27, and CD29), which are all located on (a) Illgraben fan. (b) LiDAR sensors and cameras hang over the channel and (c) are mounted to a wagon. At CD27, there is a camera looking upstream and a LiDAR scanning the central part of the channel as well as one scanning laterally to cover the entire channel cross section close to the CD.

vidual features, such as boulders and wood. However, as summarized above, an understanding of individual features and surge trajectories is necessary to improve our understanding of the mechanisms governing debris-flow motion. Earlier approaches in obtaining such data have relied on cameras and scale references with significant uncertainties and could only track a very limited amount of objects (~30 boulders) [26], [27].

The lack of direct field data has further limited our ability to understand key scaling properties of debris-flow surges, as well as how SWs and individual features interact. In particular, observations on the wave-breaking behavior could help explain the energy dissipation mechanisms and destructive potential [28]. However, no criteria for this exist due to a lack of data, but similarly to ocean waves, a breaker index may be applicable [28]. Additionally, while scaling relationships between flow depth and wave velocity have been established in hydraulics literature [29], they remain unvalidated for debris-flow waves [21]. Finally, the interaction between SWs and features on the surface of the flow has never been studied, and it remains unclear to what extent surges can accelerate large boulders in the flow, potentially increasing damage potential.

Here, we address this knowledge gap by leveraging a unique dataset collected from high-resolution LiDARs and cameras, which record data from moving debris flows [30]. Individual features, such as wood and boulders, as well as SW crests, can be resolved in high resolution with this data, meaning that it has the potential to provide a foundation for a new understanding of debris-flow mechanisms. However, the large volume of data generated by these sensors (~1 TB per event) means that manual identification and tracking of objects results in very limited and incomplete datasets. To overcome this limitation, we use two off-the-shelf neural-network-based detection algorithms (YOLOv5 and v8) to identify and track debris-flow SWs and smaller features such as boulders and wood on camera images. Subsequently, we leverage our unique multimodal sensor array to project the detections onto our LiDAR point clouds to obtain 3-D position and velocity of hundreds of individual moving objects. This processing pipeline provides a unique dataset for understanding debris-flow motion and represents a novel application of fused LiDAR and camera setups. Finally, our automatic detection and tracking algorithms provide the basis for building a general detector to be applied to different events and locations.

II. Study Site and Monitoring Setup

We analyze data from two debris flows which occurred at Illgraben (46.27N, 7.61E), a debris-flow catchment located in the Rhône Valley in western Switzerland [Fig. 1(a)]. Illgraben is one of the most active debris-flow catchments in the Alps,

as it produces on average 4 to 5 events every year [31]. The Illgraben channel has been stabilized with a series of concrete check dams (CDs), which reduce debris-flow hazard to the nearby community of Susten by preventing channel migration [Fig. 1(b)]. The Swiss Federal Institute for Forest, Snow and Landscape Research (WSL) has been monitoring the site systematically since the year 2000 with instrumentation that is installed on or near these CDs. The sensors used at the Illgraben include geophones, depth sensors, and a force plate [13], [32]. Recently, an array of LiDAR sensors and video cameras has been installed along the channel [30], [33], [34]. For this study, data were collected at three monitoring stations, which are all located along the Illgraben channel and with LiDAR scanners and cameras hanging ~6 to 10 m above the channel bed (Fig. 1). All stations consist of an upstream-looking camera and one LiDAR scanning the central part of the channel. At CD27, there is additionally a laterally scanning LiDAR to increase the field of view and cover the entire cross section immediately under the scanner. Ouster LiDAR scanners (OS0 and OS1) with a sampling frequency of 10 Hz and AXIS P1448-LE cameras were deployed in this setup.

We study data collected from two debris flows, which occurred on September 21, 2021 and June 5, 2022. For the first event (see also [30], [33]), only the station CDG was operational. In this event, the debris flow front had a velocity of 0.8–1.2 m/s, and the entire event lasted for about 30 min, before the flow transitioned to a flood. About 7 min into the event, a surge arrived at the measurement location, which increased velocities by up to 1 m/s. This second surge marked a change in the velocity profile because the velocity ratio of wood and rolling boulders (RBs) decreased from ~1 to 0.6–0.7, indicating a shift from a uniform velocity profile to one with internal shear (i.e., velocity decreases with depth) [30].

In 2022, two further LiDAR-camera setups were installed at two downstream locations, referred to here as CD27 and CD29 (see also [33], [35]). In the present work, we analyze an event that occurred on 5 June. This event arrived at CDG with a front velocity of 5.5 m/s, and later at CD27 with 3.4 m/s and CD29 with 2.8 m/s [36]. More than 60 SWs followed after the arrival of the front. These waves were only visible at the stations CD27 and CD29, which means that they only developed in the lower parts of the channel between CDG and CD27. Therefore, we study SWs of this event at CD27 and CD29. Smaller features were better visible at the upstream stations, which is why these are studied at CDG for the first event and at CD27 for the second event. The camera at CDG had raindrops on the lens during the second event. Although features can still be detected on these images, we excluded them to not biasing the analysis by such artifacts.

III. Methods

A. Deep-Learning-Based Object Detection and Tracking

We test and compare the widely used object-detection models YOLOv5 and YOLOv8 [37], [38]. As single-stage detectors, the YOLO models are fast and often considered for (real-time) applications in various disciplines, such as in the medical and food industry, and have also been applied for debris-flow front detection [39], [40], [41], [42], [43]. Here, we use extra-large versions YOLOv5x and YOLOv8x, which have the highest complexity but usually provide the highest accuracy. All trainings and detections were performed on a GPU node (Graphics Processing Unit, NVIDIA RTX 4090) of the ETH computing cluster. Following the detection of individual video frames, we use the SORT algorithm [44] to establish tracks of individual debris-flow features and surges. For more details on the YOLO and SORT models, we refer to Appendix B.

B. Training Datasets and Configurations

We produced labeled datasets using Label Studio, an open source labeling platform [45]. All labeling tasks were conducted or supervised by experts, i.e., by people with significant domain knowledge. Here, the term "label" describes a rectangular bounding box placed around an object of interest by a human expert. The first dataset contains three classes of smaller debris-flow features: boulders, RBs, and wood [Fig. 2(a)–(c)]. We labeled on average every ~55th image, whereas the frequency ranged from approx. every 25th (i.e., every 2.5 s) to approx. every 200th image (i.e., every 20 s) from the arrival of the front to the end of the recording and reflects the decreasing concentration of visible features throughout the event. This resulted in a dataset containing 329 images and 1973 labels for CD27 and 87 images and 3801 labels for CDG (Table I). We recognize that the size of our labeled datasets is small compared to other deep-learning applications or benchmarks; we note that the scarcity of labels is rooted in the overall low frequency of debris-flow events and the very recent development of combined camera-LiDAR monitoring.

The second dataset contains three classes, where we differentiate between surges as diffuse, breaking, and hybrid. We define a hybrid SW as a combination of a breaking and diffuse wave, consisting of a wave that has not yet started breaking in the middle of the channel, but is breaking on the sides closer to the channel bank [Fig. 2(d)–(f)]. These classes represent different states of wave evolution [46] and were classified based on the video recordings. For the frames containing surges, every fifth frame was labeled. We also added a few negative images (i.e., not containing any waves) such that the network would learn to identify surges that were only subtle, as well as not classify the front as an SW. This resulted in 917 and 908 images with labeled SWs at CD27 and CD29, respectively (Table I). These are not unique waves, but ~60 waves labeled at different points in time.

Both YOLOv5 and YOLOv8 come with a set of close to 30 training-related hyperparameters, such as the learning rate, batch size, and data augmentation. We set the optimizer to "Stochastic Gradient Descent" (SGD). SDG is the default for YOLOv5, while ADAM is the default for YOLOv8, but for both models, we found SDG to yield better results. For the initial learning rate, we performed a sensitivity analysis and found 0.001 to perform best in terms of mAP (Fig. A1). The initial learning rate is considered one of the most influential hyperparameters [37], [38]. For the batch size used for

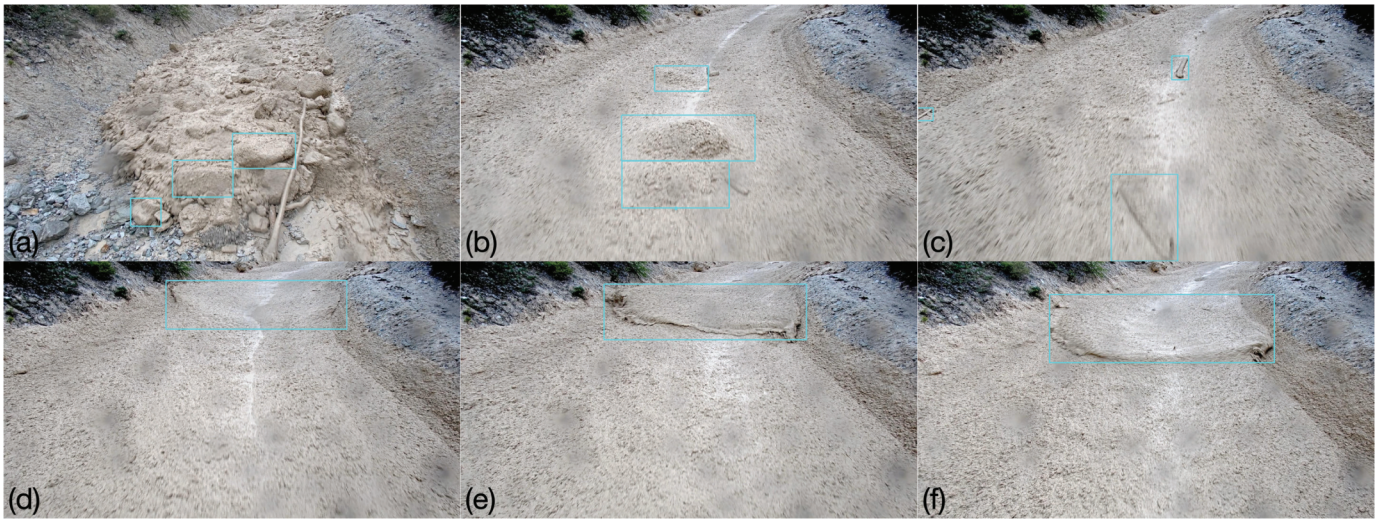


Fig. 2. Examples of debris-flow features labeled on optical images (a) boulders, (b) RBs, (c) wood, (d) diffuse wave, (e) breaking wave, and (f) hybrid wave. Note that for the reason of better visibility, only a subset of the boulder labels in (a) is shown while over 90 labels were produced in this frame.

TABLE I

OVERVIEW OF TRAINED DETECTORS. IMAGES NOT USED IN TRAINING OR TESTING WERE USED FOR VALIDATION DURING TRAINING. EACH DETECTOR DOES NOT USE A UNIQUE DATASET. FOR EXAMPLE, #3 USES COMBINED TRAINING DATA ALSO USED IN #1 AND #2. "3-D TRACKING" REFERS TO TRACKING ON OPTICAL IMAGES AND OBTAINING 3-D TRACKS BY LiDAR-CAMERA FUSION. #2, #3, AND #6 WERE TRAINED TEN TIMES WHILE PROGRESSIVELY INCREASING THE TRAINING DATASET

#	Detector objects	# Images	# Labels	Training data set	Testing data set	Use case
1	Features	329	1973	CD27 (60%)	CD27 (20%)	YOLO v5 vs. v8, 3D tracking
2	Features	87	3801	CDG (1-60%)	CDG (100-20%)	Generalizability test (single domain)
3	Features	416	5774	CD27 (60 %) + CDG (0-60%)	CDG (100-20%)	Generalizability test (multi domain)
4	Surge waves	1128	918	CD27 (60%)	CD27 (20%)	YOLO v5 vs. v8, 3D tracking
5	Surge waves	1128	918	CD27 60% of entire waves	CD27 20% of entire waves	Generalizbility test
6	Surge waves	909	908	CD29 (0-60%)	CD29 (100-20%)	Generalizbility test (single domain)
7	Surge waves	2037	1826	CD27 + CD29 (0-60%)	CD29 (100-20%)	Generalizbility test (multi domain)

training, we considered batch sizes of 1 to 3, whereas the largest batch sizes yielded the best results (Fig. A2). Larger batch sizes would require either downsampling of the image resolution or image cropping, which would not be favorable for our application. For the remaining hyperparameters, the default settings were used, which have been optimized for a benchmark dataset and include image augmentation like left-right flipping, image brightness, contrast, and hue changes, and mosaic, but not image rotation, among others, as these were not meaningful to debris-flow images (e.g., a debris-flow will never occur upside-down). When turning these augmentations off for training, the mAP dropped from around 0.7 to 0.6. For nonmaximum suppression, we used an IoU threshold of 0.45 and a confidence threshold of 0.25. The training was set for 300 epochs and the full image size of 1920 pixels, which required a low batch size of 3 due to the memory limitations of the used NVIDIA RTX 4090. With the infrastructure and hyperparameters used herein, model training for YOLOv5 and YOLOv8 took approximately equally long (2.75 h).

C. Training Methodology, Testing Datasets, and Generalizability

The labeled datasets mentioned above were randomly split into shares of 60%, 20%, and 20% images for training, validation, and testing the detectors, respectively (Table I). This widely used splitting approach quantifies the detection performance for surges and objects of the studied event at the given location; however, it cannot account for the possibility that the detector may be overfit to the specific event and location. Furthermore, especially for the SWs, information leakage from the training to the validation and testing sets is likely to skew the estimated model performance, as many labels exist for the same wave (at slightly different positions in the channel). We therefore tested a variety of data splitting techniques in order to better quantify model generalizability. In particular, we tested: 1) the generalization of the location, i.e., if a detector trained at one location along the channel has predictive power also at another downstream or upstream location; 2) if a detector trained with data from two CDs improves the predictions at one CD (multidomain learning); and 3) how many labeled images need to be added to a multidomain detector to achieve similar detection performance as a local detector.

To test the ability of the detector to generalize, YOLOv5 was additionally trained and tested with datasets from an upstream location for a different event (small features (SFs), #2 and #3 in Table I) and for downstream location for the same event (SWs, #6 and #7). Thereby, the detectors are tested with images from cameras placed at different heights and angles relative to the channel. Images were progressively added to the training starting from the model trained for CD27 (multidomain). As a reference, the performance was compared to a local detector (single-domain) to which only images from the new location were added. To test if the SW detector was

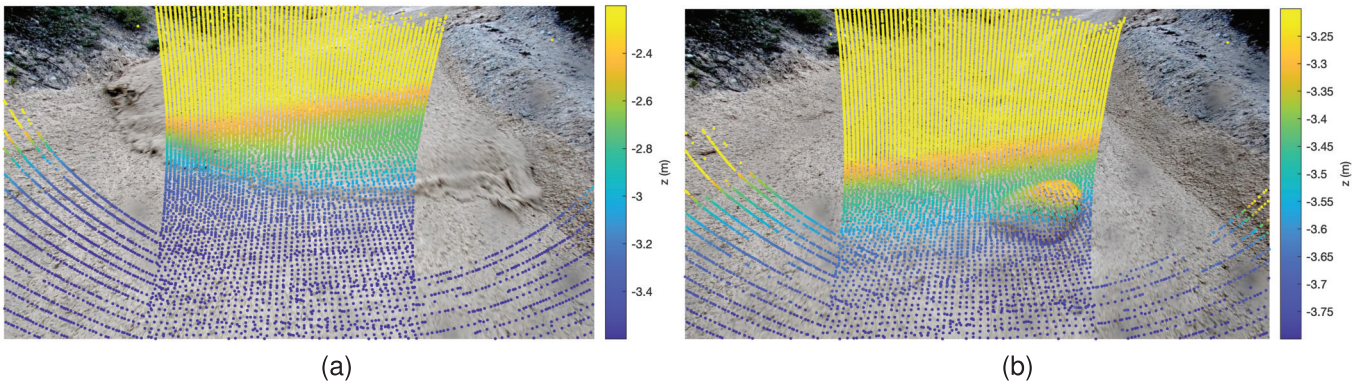


Fig. 3. LiDAR-camera fusion exemplified with (a) SW and (b) RB. The LiDAR points projected onto the image are colored according to the measurement in the z-axis. Note that the color range was set to best exemplify the alignment. Please refer to the online version of this article for color.

affected by the information leakage to the test set, a detector was trained and tested on the entire waves (#5).

We then trained seven different detectors using the different dataset subdivisions detailed above. We used the optimal set of hyperparameters determined based on the description above, and assessed the performance using: precision (P), recall (R), $F1$ score, and mean average precision (mAP) (see Appendix C). R measures how many of the objects of interest in the test set were detected by the model. P measures how many of the predicted detections are correct detections. The $F1$ score is useful to determine the optimum confidence score, i.e., the score that provides the best compromise between R and P. mAP is the integration of the P-R curve and therefore summarizes the classifiers' performance over the full range of confidence scores.

D. Camera-LiDAR Fusion

A unique aspect of the presented monitoring setup is that both LiDAR and camera data are acquired of the same scene. Such data has not been available previously for debris flows. By fusing these two data sources, we can take advantage of accurate feature detection on the relatively high-resolution camera imagery and then extract associated 3-D information from the LiDAR point clouds. Fusing the camera image and the LiDAR point cloud requires that the camera image is calibrated (i.e., corrected for lens distortion), and that the rigid body transformation (i.e., rotation matrix and translation) between the LiDAR and camera is known. Small errors in the transformation may lead to substantial uncertainties. For example, an angular error of 0.5°, which is a conservative error estimate [47], leads to an uncertainty of about 2.6 to 17.4 cm within a distance of 3 to 20 m, which is a typical range in this study.

We adapted a target-based method for the rigid body transformation [48]. The approach relies on the automated identification of a target (checkerboard) in a large number of corresponding camera images and point clouds. This procedure results in good alignment, as exemplified in Fig. 3, where surface elevation is shown to clearly increase along surface features such as roll waves and boulders when the LiDAR points are projected onto the camera image. Due to a bend in the channel, the flow is expected to elevate on the right channel side, which is also detectable in the colored point clouds [Fig. 3(a)]. The camera-LiDAR fusion is used to project the object detection results from the images onto the point clouds (Fig. 4). Hence, 3-D information is available for all detected objects and allows for determining geometric properties.

E. Derivation of Geometric and Kinematic Properties of Debris-Flow Objects

The object detections in combination with the tracking algorithm, which assigns a trackable object a unique ID in independent images, were used to derive object velocities. To this end, we used the 2-D bounding box midpoint to calculate the velocity of boulders and wood, while for SWs we used the 2-D midpoint of the lower bounding box boundary, as the fronts of the SWs were detected with higher precision than the entire wave [Figs. 2(a) and (b) and 4(d)]. These tracks were validated with tracks produced manually and compared to surface velocities produced by [36].

SW height was derived by calculating the difference between LiDAR points on the wave crest and the LiDAR points at the same location before the arrival of the debris flow. Due to the CDs stabilizing the channel, only a little erosion and deposition is expected. The crest location is derived from cross-sectional LiDAR data [Fig. 3(d)]. As the crest represents the maximum curvature, it is identified with the maximum in the second derivative along the y-axis.

Other properties related to SW propagation are the depth of the fluid downstream of the SW, which was extracted at the same location as the crest, but in a point cloud 25 seconds before the crest arrived. Likewise, the velocity of the downstream fluid could be extracted from the PIV-derived surface velocity [36]. Hence, the derived SW velocities can be compared to theoretical solitary wave propagation [29]

$$c = \sqrt{gh_2} \tag{1}$$

where c is the wave celerity, g the gravitational constant, and h_2 is the flow depth at the wave crest. c can be approximated by $v_w - v_1$, where v_w is the wave velocity and v_1 is the velocity of the undisturbed fluid downstream of the wave. Furthermore, a wave breaker index H_b/h_b can be calculated analogously to

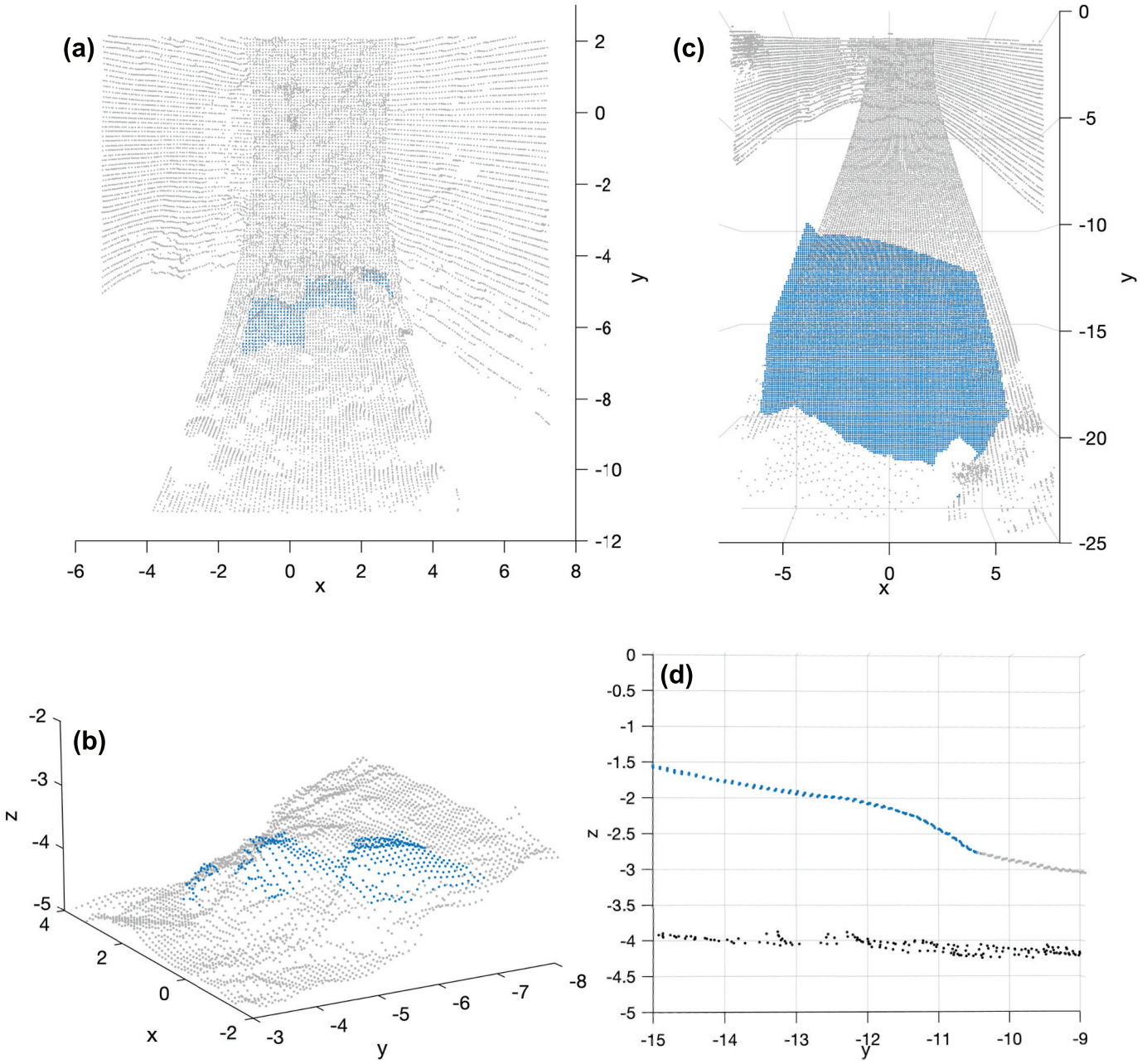


Fig. 4. Bounding boxes (blue) projected onto the LiDAR point clouds (gray). The bounding boxes of boulders in Fig. 2(a) are shown from (a) above and (b) side. The breaking wave in Fig. 2(e) is shown from (c) above and (d) as a cross section in the center of the channel, including the pre-event channel bed topography (black). Please refer to the online supplementary for a movie of the moving boulders (Movie S3).

as is established for ocean waves [28], where H_b is the wave height ($h_2 - h_1$) and h_b is the depth at the point of breaking, which we estimated as the mean of h_2 and h_1.

IV. Results

A. Object Detection

YOLOv5 and YOLOv8 were compared for the detection of both small debris-flow features (boulders, RBs, and wood) and different SW types (diffuse, breaking, hybrid) (Table II). We find that both YOLOv5 and YOLOv8 reach mAPs of around 0.7 for SFs and almost 1.0 for SWs, whereas YOLOv5 performs slightly better for the SFs. The similar model performances are also visible in the mAP50-95 as well as in the PR curves (Fig. 5). Based on these findings, YOLOv5 was used to generate the results for all the following analyses.

The generalizability tests showed that while the performance of the single-domain detector keeps increasing until the maximum of 60% of available images are added to the training, the performance of the multidomain detector converges already between 16% and 24% of added images (Fig. 6). At a maximum of 60% of added images, the two detectors show similar performance. Analogously, a test was conducted for the SW detection (Fig. A3). The multidomain detector for SWs had the same but smaller effect as for smaller objects. However, the analysis also showed that both

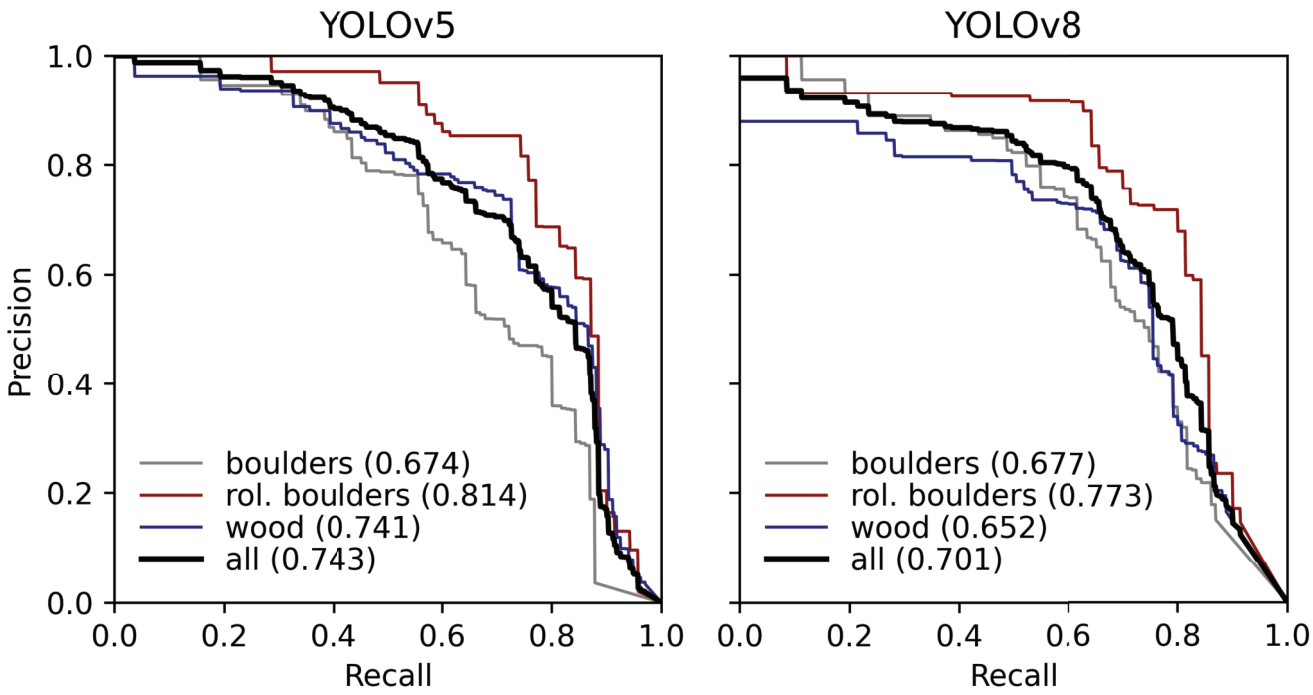


Fig. 5. Precision-recall curves for YOLOv5 and YOLOv8 computed for the test set. In parentheses, the AP of individual classes is given as well as the mAP over all classes (see also Table II).

TABLE II

PERFORMANCE METRICS ON THE TEST DATASET OF A DETECTOR FOR SFS AND FOR SW IN DEBRIS FLOWS FOR YOLOV5 (V5) AND YOLOV8 (V8). THE METRICS SHOWN ARE PRECISION (P), RECALL (R), $F1$ SCORE, MAP FOR AN INTERSECTION OVER UNION IOU = 0.5 (MAP50) AND FOR THE MEAN WHEN IOU RANGES BETWEEN 0.5 AND 0.95 (MAP50-95)

Model	Class	# Images	# Instances	P		R		F1		mAP50		mAP50-95	
				v5	v8	v5	v8	v5	v8	v5	v8	v5	v8
SF	all	66	320	0.74	0.71	0.69	0.67	0.71	0.69	0.74	0.70	0.39	0.39
	boulder	10	115	0.66	0.66	0.60	0.65	0.63	0.66	0.67	0.68	0.34	0.39
	rollingboulder	37	70	0.81	0.76	0.76	0.70	0.78	0.73	0.81	0.77	0.44	0.41
	wood	49	135	0.74	0.71	0.72	0.64	0.73	0.68	0.74	0.65	0.38	0.37
SW	all	226	185	0.98	0.97	0.97	0.96	0.97	0.96	0.98	0.98	0.70	0.67
	diffuse	239	128	0.97	0.94	0.95	0.98	0.96	0.96	0.98	0.97	0.63	0.62
	breaking	39	39	0.98	0.98	1.00	1.00	0.99	0.99	1.00	1.00	0.73	0.71
	hybrid	18	18	1.00	1.00	0.94	0.89	0.97	0.94	0.96	0.99	0.72	0.69

single- and multidomain detectors converge at <20% of labels. Finally, a detector trained and tested on a dataset split by entire waves showed that the performance only decreased marginally (Table A1).

B. Tracked Objects Velocities

Frame-by-frame tracks were established using the SORT algorithm (Movies S1 and S2). LiDAR-camera fusion allowed for projecting the tracks onto the point clouds (Movie S3) to obtain object velocities from the 3-D information. Only tracks lasting over more than 10 frames (i.e., 1 s) and with a minimum mean average velocity of 1 m/s were considered. In doing so, we ensured that the mean velocities were calculated over a representative stretch and that deposited objects were excluded. This resulted in almost 1000 unique features in the analyzed event [Fig. 7(a)–(c)]. Boulders were mainly detected in the beginning of the event, while the RB occurrence peaks later and still appears in the end. Wood is the most detected feature and occurs throughout the event. The velocity tracks show that the wood tends to be faster than (rolling) boulders, especially in the early event phases, and that the velocity decreases with event time. Most SWs were of the diffuse type [Fig. 7(d)–(f)]. In total, 44 SWs were tracked, and they occurred throughout the event.

The velocities inferred from the automatically tracked objects can be compared to the velocities of manually tracked features and SWs (Fig. 8, [36]). Manually labeled and automatically detected boulders have overlapping clusters during the passage of the front and the first minutes. For wood and RBs, certain objects were tracked with similar velocity and fall on the surface velocity line (Figs. 8 and A4), whereas wood tends to be faster than RBs. Manually and automatically

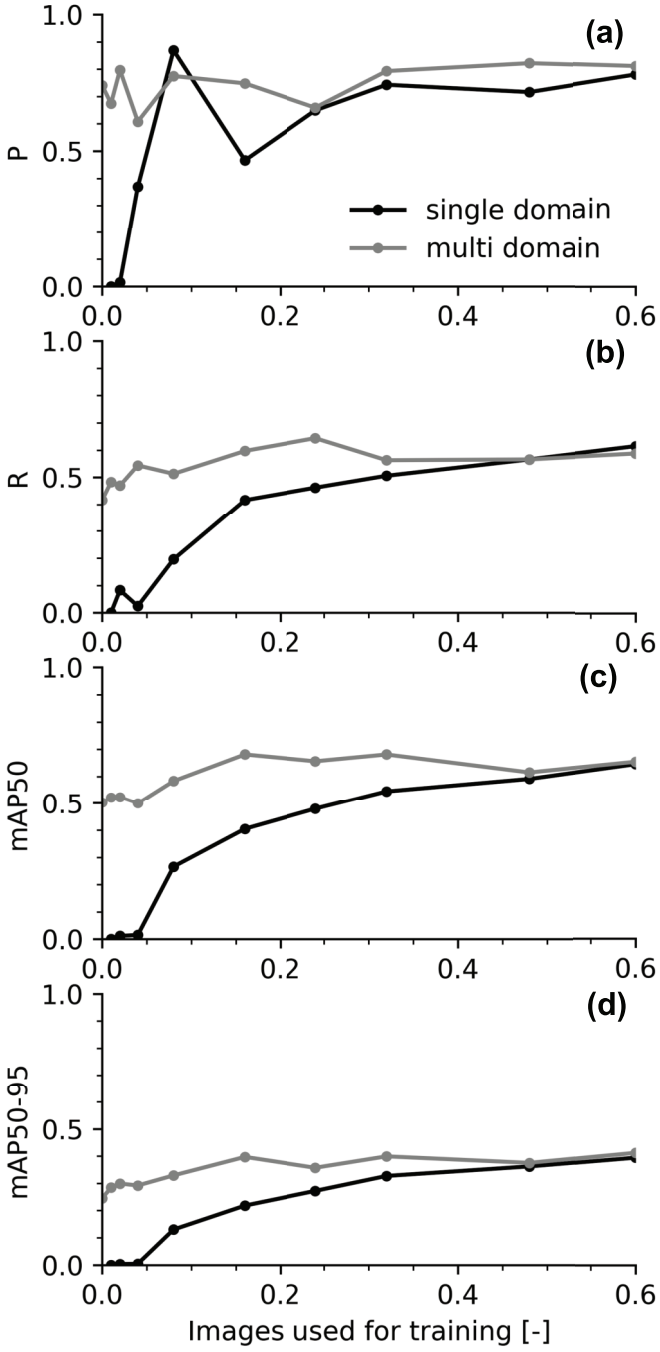


Fig. 6. (a) Precision, (b) recall, and (c) and (d) mean average precision for feature detectors trained and tested against the dataset from CDG, with images progressively being added to the training based on the CD27 model (multidomain) and one trained only locally without prior fine-tuning (single domain).

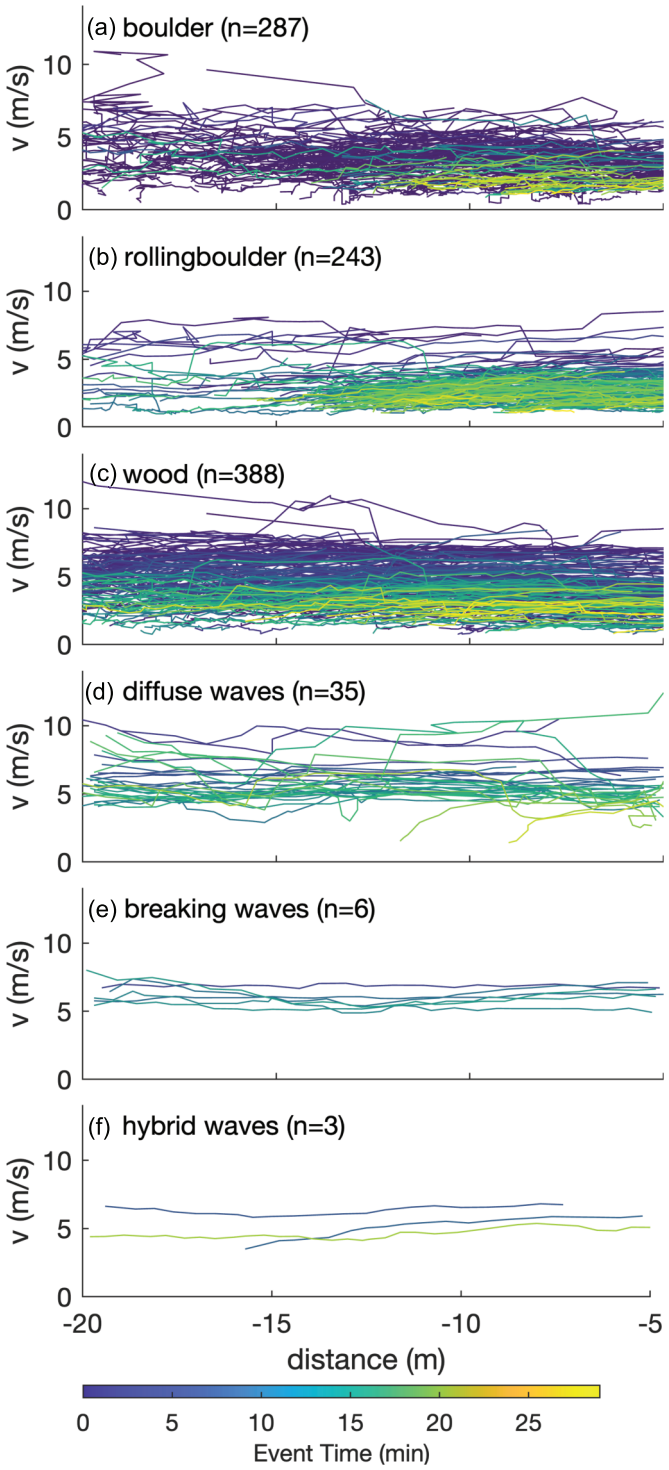


Fig. 7. Velocities of detected and tracked objects from the feature model and the surge-wave model, colored according to the time of their occurrence in the event: (a) boulders, (b) RBs, (c) wood, (d) diffuse waves, (e) breaking waves, and (f) hybrid waves. The *x*-axis refer. The velocity tracks were smoothed with a moving mean with a size of 10 (i.e., 1 s). Please refer to the online version of this article for color.

tracked surge waves result in very similar velocities, which are higher than the surface velocity. Overall, many of the manually and automatically derived tracks overlap or are very close, thereby validating the automatic approach.

The generated wave tracks show that the waves grow in height as they move downstream [Fig. 9(a)]. The solitary wave propagation is well estimated by the theoretical line for waves at CD29, but underestimated for waves at CD27 [Fig. 9(b)]. Breaking and diffuse waves are well separated by a breaker index threshold of 0.4, below which there is only one breaking wave and 88% of the diffuse waves [Fig. 9(c)]. The few hybrid

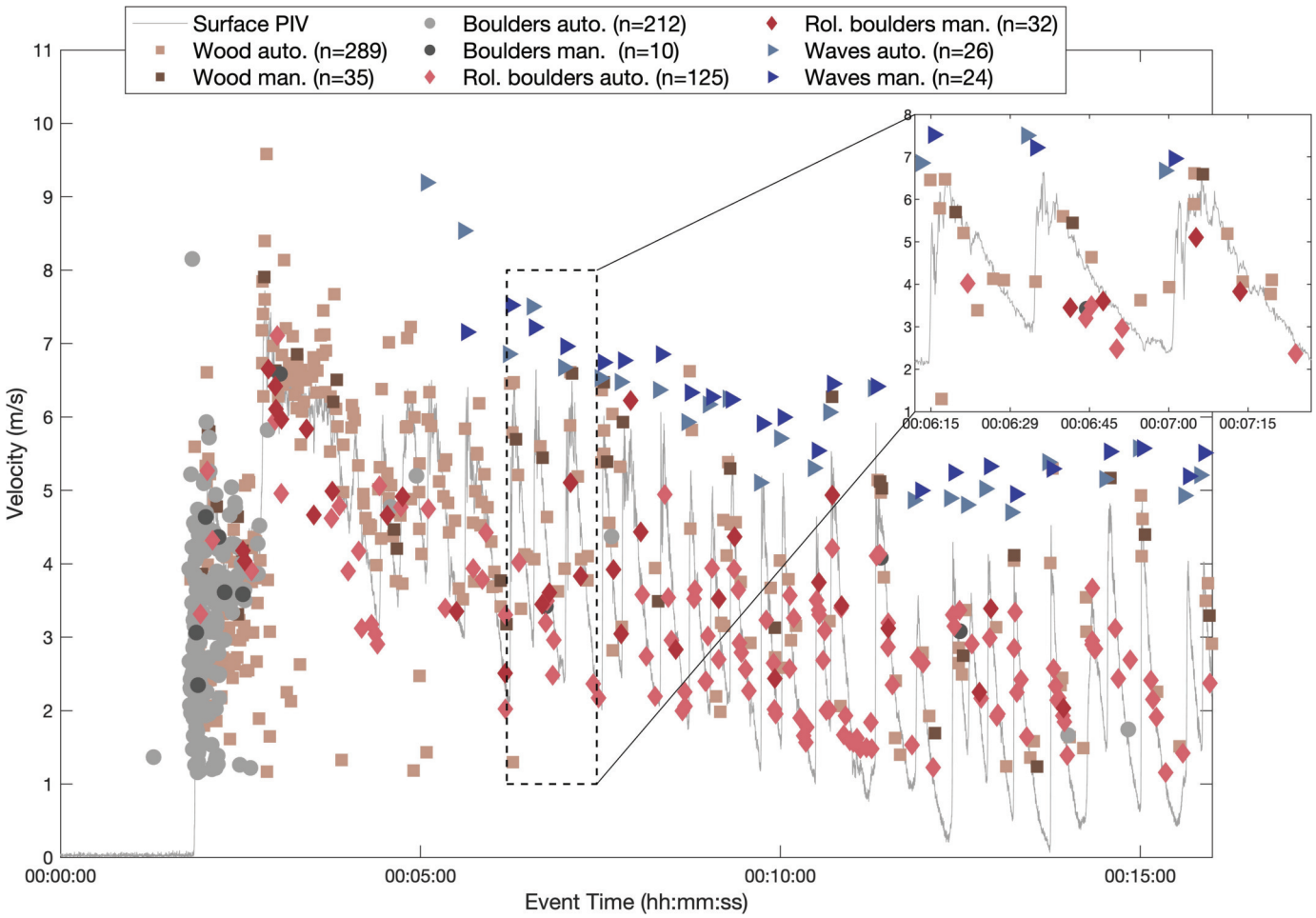


Fig. 8. Mean feature, wave, and surface velocity. Manually derived velocities and the surface velocity were produced by [36]. Please refer to the online version of this article for color.

waves are all very close to this threshold, and no waves were observed above a breaker index of 0.6.

Combining the feature and surge wave detector allows for analyzing the wave-feature interaction. We filtered tracked features that also had waves being detected in the same frame. By splitting the tracks according to their position relative to the waves (in front or behind), it becomes evident that the RBs and wood in front of the wave are slower and are sped up by the waves (Fig. 10). Since boulders mostly occur at the front and immediately after, there were no visible interactions between boulders and waves.

V. Discussion

We have presented the detailed analysis of a novel dataset collected from moving debris flows, based on high-resolution LiDARs and cameras [30]. Detailed processing of this data requires automated methods, and we implemented a novel processing pipeline to detect debris-flow objects on 2-D camera images and subsequent fusion with LiDAR point clouds to obtain 3-D information. We obtained almost 1000 tracks for one event and one location, which is more than 300× more than from earlier manual approaches. The presented detectors and subsequent processing provide the basis for efficient data processing at other locations and for other events to obtain tracks of boulders, wood, and surge waves in 3-D.

A. Performance of Object Detectors

For object detection, YOLOv5 yielded slightly better results than YOLOv8 in terms of commonly used metrics such as the mAP. This is surprising, as YOLOv8 has been shown to be more accurate in independent comparative studies, e.g., [38], [49]. However, a limitation of our comparison is that we did not explore the full hyperparameter space but only varied the initial learning rate, batch size, and optimizer. It is possible that the YOLOv8 performance would increase if all hyperparameters were to be optimized individually. This possibility was not explored for the following reasons: 1) a full hyperparameter evolution comes with high-computational costs; 2) the initial learning rate, batch size, and optimizer are often considered as the most influential hyperparameters but the results only changed marginally within a wide range of hyperparameter values (Figs. A1 and A2); and 3) our dataset is relatively small compared to benchmark datasets commonly used for model architecture comparisons, such that the resulting difference we see in YOLOv5 and YOLOv8 can likely be attributed to input data rather than to the

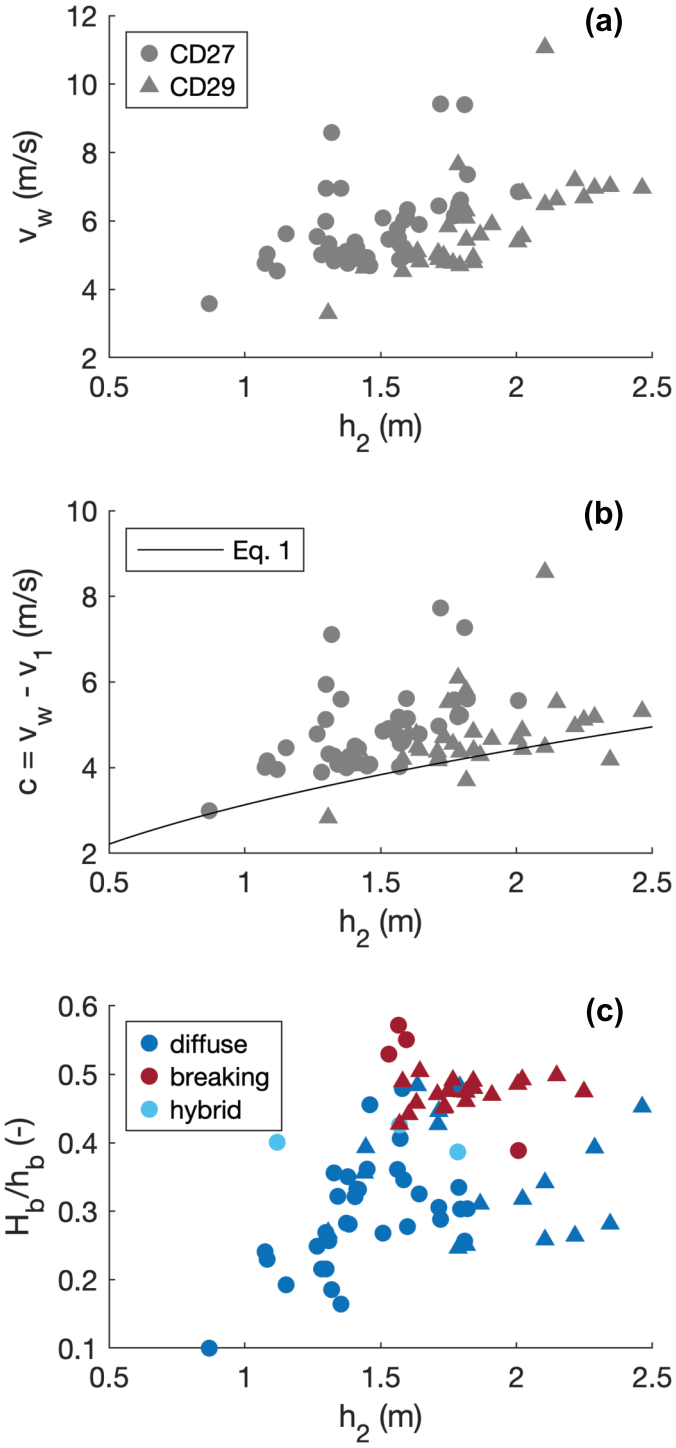


Fig. 9. (a) Wave velocities, (b) measured and theoretical celerity, and (c) amplitude ratios for detected surge waves at CD27 and CD29 as a function of the flow depth at the wave crest (h_2).

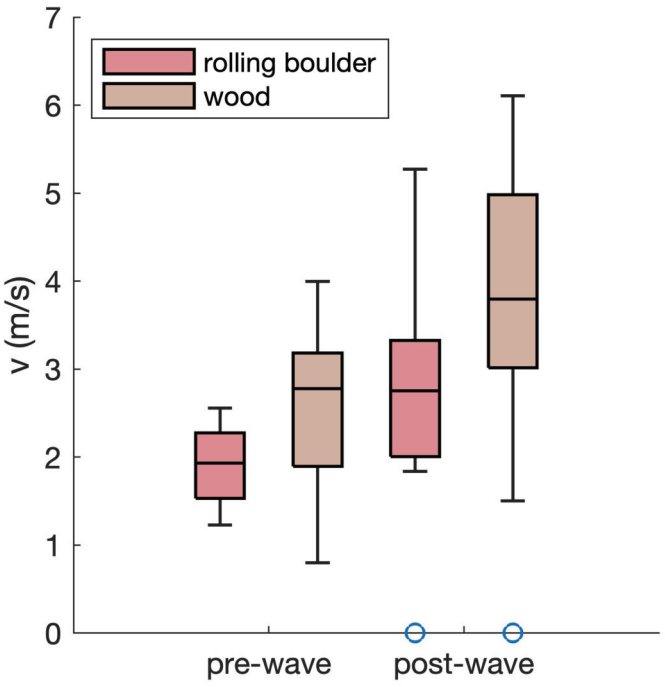


Fig. 10. Mean velocities of features (RB: rolling boulders, W: wood) detected in frames where also surge waves were detected. The classes were split into detections before (pre) and after (post) they were overtaken by a surge wave. Please refer to the online version of this article for color.

model architecture. This would be one possible explanation for why other comparative studies likewise found YOLOv5 to outperform YOLOv8, e.g., [50], [51].

Although we pursued the following analysis with YOLOv5 to obtain the highest possible detection accuracy, YOLOv8 may be reconsidered in the future if more emphasis is put on the real-time operation (e.g., for early warning purposes). Such an adjustment would, in our case, speed up the detection by five times or more at the cost of a minimal drop in P and R (Table II). We note that careful investigations are required to determine whether more accurate but slower or less accurate but faster data processing delivers more reliable and timely early-warning results.

Boulders and wood can be much smaller than the waves and subject to occlusion. Both these factors affect the detection accuracy [52] and are reflected in the performance metrics, whereas the surge wave detector performs better than the feature detector (Table II). Additionally, feature rotation and temporary submerging are expected to affect the detection and tracking. This is also evident in the recall metric (0.69), indicating that a substantial amount of features is missed in the automated detection. As expected, the boulder class, which has some of the smallest features, had the lowest recall (0.60). This is, however, not considered to affect the interpretation of track velocities, as the temporal structure of object occurrence (Fig. 7) matches observations from videos and process understanding (e.g., bouldery debris-flow front). More importantly, though, is that the precision metric (0.74) can be considered as good, i.e., the number of falsely classified features is low. Furthermore, a falsely classified object from the background is unlikely to be tracked and should therefore not be represented in the analysis shown here. Therefore, the tracking algorithm can be seen as a filter for false detections.

B. Object Detector Transferability

Despite the conventional procedure of splitting all labeled data into training, validation, and test datasets, the question remains how transferable the detectors are to other events and

other locations. We showed that by applying the detector to other locations along the channel, and to another event for the case of SFs, less than 25% of the images available for the respective dataset had to be added to the training to perform as well as a local detector (Figs. 6 and A3). However, the performance metrics of the multidomain and local detectors were similar, suggesting that using a pretrained network can decrease the amount of training data needed to apply it at a new location, but that its performance in terms of precision and recall will ultimately be similar. This result is in line with findings for multidomain rockfall detectors [53]. These findings are limited by the analysis of only two debris-flow events from a single catchment. However, data from recent events and new monitoring stations offer an opportunity to validate these results and explore multidomain learning for a global or alpine debris-flow detector. The observed transfer learning across differently aligned cameras—though within the same catchment—suggests the potential for developing a more generalizable model.

Each dataset took us about two days to produce (per location and dataset type). Hence, for inference of objects in future events, 1.5 working days or about 12 h will be saved, and we expect more time to be saved with labels from future events being added over time. It should be noted that this is time saved compared to producing labels for an off-the-shelf model. The time saved by automatically detecting and tracking objects amounts to ~250 h per event (assuming 15 min working time per manual track).

C. Geometric and Kinematic Object Properties

The velocities calculated from the tracks agree with velocities of features and waves derived manually (Fig. 8). This is evidence of a high bounding box accuracy and justifies that the velocities can be derived from the bounding box midpoint and the lower boundary midpoint for SFs and waves, respectively. There are some boulders with low velocities before the arrival of the front. This can be explained by the detection of static boulders in the channel, which were then not immediately covered by the front but entrained and tracked. The calculated mean velocity of these features is then naturally lower. Similarly, the wood between 3 and 5 min with lower velocity can be explained with temporary deposition on the channel bank due to a decrease in flow depth, and then re-entrainment by the subsequent surge wave. These outliers could easily be filtered if necessary, but also demonstrate the ability of meaningful detection, tracking, and velocity determination, as well as great potential for using these methods to understand longitudinal sorting and levee formation. The longitudinal sorting process is thought to be linked to the characteristics of the vertical velocity profile present in the flow, as this profile drives the preferential transport of large boulders to the front [15]. However, the shape and dynamics of velocity profiles in debris flows is understudied since it is very difficult to measure. The only field measurements available indicate that debris flows have varying flow regimes during an event and also from event to event [54]. An alternative but indirect way of interpreting the velocity profile is comparing the velocities of individual objects floating on the surface, such as wood, and larger boulders with the approximate size of the flow depth, as it is thought that the floating objects represent the velocity at the surface while the velocity of the larger boulders are influenced by the entire velocity profile [30]. Therefore, this study provides a basis for tackling research questions related to the vertical velocity profile [30], [54], and the motion of individual boulders at the front [14].

We showed that surge waves grow as they propagate downstream [Fig. 9(a)], which is in line with the theoretical understanding of surge wave propagation [23]. Furthermore, waves have a strong tendency to break at a breaker index threshold of 0.4, and we observed no waves above 0.6 [Fig. 9(c)]. This may represent an upper limit for surge wave heights, but it has to be confirmed with more events. Herewith, we have provided the foundation to further explore these phenomena.

The feature and surge wave detectors can be used in combination to study how features are affected by the surge waves. The results suggest a significant acceleration of features by waves (Fig. 10). To validate this approach, we split the tracks in the middle and calculated feature velocities upstream and downstream of the midpoint. Upstream and downstream velocities are very similar and not significantly different, proving that the acceleration is indeed through the surge wave and not a general acceleration in the channel (Fig. A6). This provides the basis for studying the mobility of such surge waves. For example, a piece of wood at the flow surface represents the movement of a particle at the flow surface during the passage of a surge wave and could be compared to the theory of particle trajectories in gravity waves [55]. Furthermore, hypotheses on surge wave propagation could be tested. We showed that the wave celerity is close to the theory of solitary waves (Fig. 9), but underestimated at CD27. However, more complex theories, as for example a uniform progressive flow [29], [56] should be tested in the future.

Uncertainties in track velocities stem from the LiDAR-camera fusion, the LiDAR measurement, and the accuracy of the bounding boxes, both from the manually drawn boxes for training and the model detections. When averaged over 1 s, the velocity tracks do not show systematic acceleration or deceleration, although the smallest features (boulders) show the largest variations (Fig. 7). As the distance to the sensors increases, the residual velocity (deviation of instantaneous velocity from mean velocity) increases due to angular errors of the LiDAR-camera fusion and the LiDAR beam divergence, and less accurate bounding boxes. These residuals are expected to be ± 1.8 m/s at a distance of 20 m to the sensor and decrease linearly to <0.5 m/s when approaching the sensor (Fig. A5). However, as the residual velocities show no trend in the positive or negative direction, no biases are introduced in mean velocity estimates, including moving mean velocities over sufficiently large windows.

Finally, studying the velocities of individual boulders is also relevant to the design of debris-flow mitigation measures, as boulders have more localized and often higher impact than the fluid component [57], [58]. Because the velocity of individual boulders is mostly unknown, structural impact studies rely

on estimates of the bulk flow velocity, which is commonly calculated by means of open channel hydraulic theory such as Manning's formula [59]. The presented framework could therefore help develop empirical relations of bulk flow and boulder velocity to reduce uncertainties in the design of mitigation structures.

VI. Conclusion

This study demonstrated that deep-learning-based processing of novel debris-flow data can yield new insights into debris-flow mechanics. In particular, the new processing pipeline allows for extracting high-resolution data of debris-flow features such as surge waves, boulders, and wood. Since novel debris-flow monitoring techniques with cameras and LiDARs work at high temporal resolution (10 Hz), this automated processing is orders of magnitude faster than manual processing.

We showed that a multidomain detector requires significantly less new training data than a location-specific (local) detector while maintaining the performance. This implies significantly less labeling effort when adapting the detector to new locations. Interestingly, YOLOv5 performed marginally better than YOLOv8. While this performance gap could potentially be reduced through hyperparameter optimization, significant improvements are unlikely with the current training dataset. Therefore, we used YOLOv5 results for further processing. However, YOLOv8's superior speed makes it more suitable for real-time applications, especially where marginal accuracy differences are acceptable.

Automatically derived feature velocities closely matched manual tracking results. Most boulders were detected with the arrival of the unsaturated front, while RBs and wood were detected throughout the event. Wood moved faster than RBs on average, indicating the presence of internal shear within the flow [30]. Surge waves were the fastest features and were observed to accelerate wood and RBs. The dataset also enabled the calculation of a wave breaking index, adapted from coastal engineering, with wave breaking occurring at a threshold of 0.4. The detectors will be applied to produce more datasets of feature tracks, velocities, and sizes large enough to perform statistical analyses, which is critical to address key open research questions in debris-flow dynamics.

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Jacob Hirschberg was born in 1990. He received the Dr.Sc. ETH degree in environmental engineering from ETH Zurich, Zürich, Switzerland, and Swiss Federal Institute for Forest, Snow and Landscape Research (WSL), Birmensdorf, Switzerland, in 2021.

Since 2023, he has been a Lecturer and Senior Researcher at the Chair of Engineering Geology, ETH Zurich. His research interests include early warning, monitoring, and climate change impact assessment of natural hazards with a focus on debris flows and related geomorphic processes.

Valentin Tertius Bickel was born in 1990. He received the Dr.Sc. ETH degree in earth science from ETH Zurich, Zürich, Switzerland, and the Max Planck Institute for Solar System Research, Göttingen, Germany, in 2021.

He is a Research Fellow at the Center for Space and Habitability, University of Bern, Bern, Switzerland. His research interests include neural networks for object and change detection, as well as planetary geomorphology and terramechanics.

Jordan Aaron was born in 1990. He received the Ph.D. degree in geological engineering from the University of British Columbia, Vancouver, BC, Canada, in 2017.

Since 2023, he has been an Assistant Professor of Engineering Geology at ETH Zurich, Zürich, Switzerland. His research interests include numerical and laboratory modeling and field instrumentation of engineering geology systems, with a particular focus on landslides.